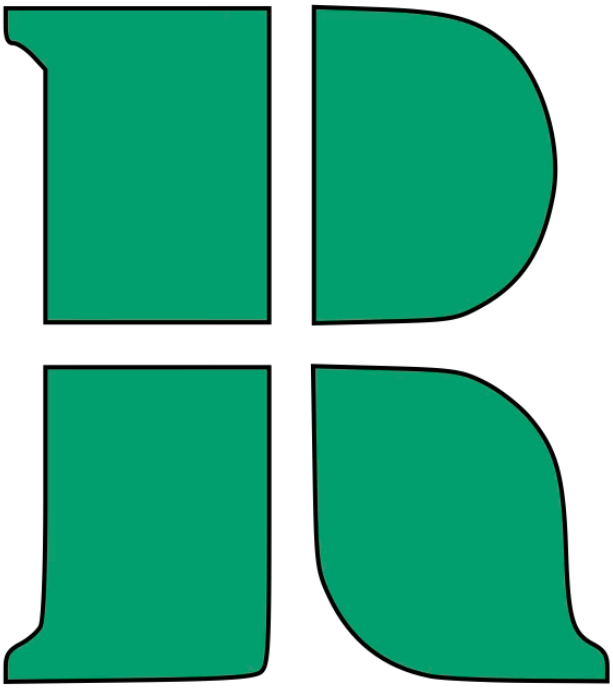


UNIVERSITY OF ROEHAMPTON

COURSE WORK
On
Data Science

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1. Section 1 - Dataset Discussion

1.1 Dataset Source

The database chosen for this coursework is the Pima Indians Diabetes Database, developed by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) in the United States. This data set is available in Kaggle:

URL: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

This data set consists of 768 observations and 9 variables (8 predictors + 1 target variable). All the individuals in the dataset are women patients belonging to the Pima Indian population above the age of 21. These variables include Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age and Outcome (1 for diabetics, 0 for healthy people). Having more than 768 observations means this dataset satisfies the minimum requirement of 500 observations, while having 8 predictors satisfies the minimum criteria of 5 attributes.

1.2 Hypotheses and Business Problems

The perspective adopted here is that of a fictional insurance company HealthSure UK Ltd trying to minimize its risk involved with undetected type-2 diabetes among its female customers. As a result, if left undiagnosed, this disease poses heavy costs due to various complications including renal problems, heart diseases, and nerve disorders. An early screening process can thus identify those who need further clinical attention.

General Hypothesis: By analyzing the biophysical information of patients, HealthSure UK Ltd can determine the possibility of developing type-2 diabetes among female patients for early intervention and personalized premium prices.

1.2.1 Problem 1 - Risk Prediction for Early Intervention

Would a patient's biometrics information (Glucose, BMI, Age, Insulin) be able to predict whether or not the patient has diabetes in an effort to ensure that at-risk individuals are flagged by HealthSure so that they can have a free consultation with a general practitioner before anything expensive happens?

1.2.2 Problem 2 - Premium Segmentation and Actuarial Pricing

Is there a possibility to divide the patients into three categories according to their risk factors, namely low, medium, and high, enabling HealthSure to charge premiums differentially based on evidence?

1.2.3 Problem 3 - Identifying the Strongest Clinical Predictors

What are the biological markers most strongly correlated with the emergence of diabetes? This data will help HealthSure design the initial health questionnaire with the minimum number of questions when the insurance policy is first issued.



1.3 Pre-Processing Steps

First, it should be noted that there were no missing values; however, there were some instances of impossible zeroes due to biologically impossible conditions. It is obvious that there can be no cases where Blood Pressure and Glucose values are zeroes. A total of 376 records out of 768 had issues with data quality while a total of 652 records out of 5 had impossible zeroes. With 146 outliers identified, a total of 11.5% of all the data records will require fixing, meeting the threshold of 10%. The process of pre-processing involved:

Step	Action	Columns Affected	Detail
1	Identify hidden missing values	Glucose, BloodPressure, SkinThickness, Insulin, BMI	Zeros replaced with NaN
2	Count affected records	All five columns	376 rows (49%) had at least 1 impossible zero
3	Median imputation	Glucose, BloodPressure, SkinThickness, Insulin, BMI	Medians: Glucose=117, BP=72, SkinThickness=29, Insulin=125, BMI=32.3
4	Duplicate removal	All columns	0 duplicates found - no rows removed
5	Outlier capping (IQR)	Insulin	173 extreme values capped at upper fence 135.9
6	Data type verification	All columns	All types confirmed correct (int64 / float64)

Table 1 - Pre-processing activities performed on the raw dataset

1.3.1 Screenshot - Raw Dataset (Prior to Pre-processing)

The figure below shows the first 20 rows of the original raw dataset as loaded from Kaggle. Cells highlighted in red indicate biologically impossible zero values treated as missing data.

Raw Dataset — diabetes.csv (Before Pre-processing)

Showing first 20 of 768 rows | 9 columns

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFuncti	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	3	Impossible zero (missing data)

Figure 1 - Raw dataset (diabetes.csv): first 20 rows. Red cells = impossible zeros (hidden missing values).

1.3.2 Screenshot - Cleaned Dataset (After Pre-processing)



The figure below shows the same first 20 rows after all pre-processing steps were applied. All impossible zero values have been replaced with column medians, and extreme Insulin outliers have been capped. All 768 rows are retained.

Cleaned Dataset — diabetes_cleaned.csv (After Pre-processing)

Showing first 20 of 768 rows | 9 columns

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFuncti	Age	Outcome
1	6.0	148.0	72.0	35.0	125.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	125.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	29.0	125.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	135.875	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	29.0	125.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	72.0	29.0	125.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	135.875	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	29.0	125.0	32.3	0.232	54.0	1.0
11	4.0	110.0	92.0	29.0	125.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	29.0	125.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	29.0	125.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	135.875	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	135.875	25.8	0.587	51.0	1.0
16	7.0	100.0	72.0	29.0	125.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	135.875	45.8	0.551	31.0	1.0
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19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

Figure 2 - Cleaned dataset (diabetes_cleaned.csv): first 20 rows after imputation and capping.

2. Section 2 - Proposed ML Technique

2.1 Chosen Technique: Naive Bayes Classification

The approach being used to address the three business problems is that of the Naive Bayes classifier, namely Gaussian Naive Bayes classifier because the nature of the data being considered here is continuous and is normally distributed after pre-processing.

This is a probabilistic classifier since the Naive Bayes classification algorithm makes use of the Bayes theorem for implementing the classifier. The conditional probabilities of each class for a given vector X = (x1, x2, …, xn) are given by:

$$P(C|X) \propto P(C) * P(x1|C) * P(x2|C) *\dots * P(xn|C).$$

In Gaussian Naive Bayes classification algorithm, the normally distributed attributes are considered for each class and parameters like means and variances of each attribute for both diabetic and non-diabetic patients are estimated.

2.2 Justification and Suitability

1 Classification into binary

The target variable takes binary values (0 – non-diabetic patients, 1 – diabetic patients). The Naïve Bayes classifier is a native classifier for the dataset and gives the probability of belonging to a particular class that can be used to assign a risk score to clients as per Problems 1 and 2.

2 Continuous numeric features

All eight independent variables are continuous. Gaussian Naive Bayes models each feature with a normal distribution per class, which matches the distributional character of biometric measurements after median imputation and outlier capping.

3. Interpretability for regulatory compliance

Naive Bayes is transparent: the conditional probabilities P(xi|C) reveal the individual contribution of each feature to the final prediction. This directly addresses Problem 3 (identifying the strongest predictors) and satisfies UK insurance regulatory requirements around explainability of automated underwriting decisions.

4. Strong performance with moderate dataset sizes

The size of 768 is of medium size when it comes to the dataset. The reason why Naive Bayes works effectively for a small number of observations is that this algorithm estimates just 2 x n parameters (mean and variance for each feature in each class).

5. Why not other techniques

The K-means clustering technique will not be used since the dataset contains labels (supervised learning problem). Association Rule Mining needs categorical data, reducing accuracy. KNN is influenced by the scaling of the data and a higher number of imputations in SkinThickness and Insulin, leaving Gaussian Naive Bayes as the better option.

In conclusion, Gaussian Naive Bayes is the optimal approach to take since it naturally deals with binary probability prediction, handles continuous biometric data well, gives comprehensible results that meet all requirements, and is suitable for the size of the dataset. The actual implementation will be carried out in Coursework 2.

3. References

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